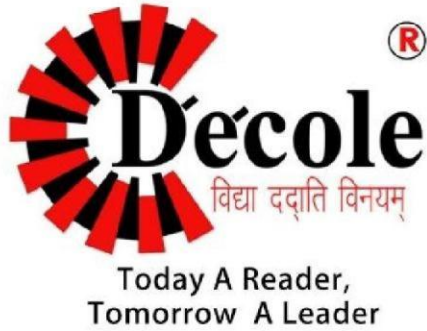




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Title of Project

Assignment File

Subject Name

Data Structure and C++

Submitted By

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Department - BCA

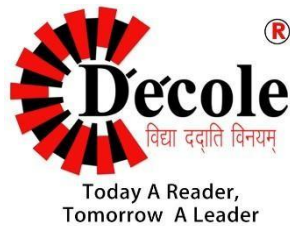
Semester – 2nd

Session – 2025-26

Project Report of
Data Structure and C++

Submitted to
Dezyne École College
Towards the Partial Fulfilment of
BCA 2nd Semester

By
Meenakshi Mali



www.dezyneecole.com

2025-2026

Acknowledgement

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We are proud to be part of Dezyne École College, an institution that truly nurtures professionalism, innovation, and purpose.

With sincere appreciation,

Meenakshi Mali

Department of BCA

Dezyne École College, Ajmer

1. WAP to input size of Queue from user and create a queue using array. And perform push, pop and traverse operations using menu driven programming and functions.

Code:

```
#include<iostream>

#include<conio.h>

using namespace std;

class Queue
{
public:
    int QUEUE[10];
};

class QueueOperation : public Queue
{
public:
    int Front, Rear, MaxSize;

    QueueOperation()
    {
        Front = Rear = -1;

        cout << "Enter Size of Queue (1-10) -- ";
        cin >> MaxSize;
```

```

if(MaxSize > 10 || MaxSize <= 0)
{
    cout << "Invalid Queue Size...";
    exit(0);
}
}

void Push()
{
    system("cls");

    cout << "Push Operation Module\n\n";

    int DATA;

    // Overflow Condition
    if(Rear == MaxSize - 1)
    {
        cout << "Queue Overflow...\n\n";
        return;
    }

    cout << "Enter Data -- ";
    cin >> DATA;

    // First Element

```

```

    if(Front == -1)
    {
        Front = Rear = 0;
    }
    else
    {
        Rear++;
    }

    QUEUE[Rear] = DATA;

    cout << "\nData Inserted Successfully...\n\n";
}

void Pop()
{
    system("cls");

    cout << "Pop Operation Module\n\n";

    // Underflow Condition
    if(Front == -1)
    {
        cout << "Queue Underflow...\n\n";
        return;
    }

```

```
cout << "Deleted Element -- " << QUEUE[Front] << endl;

// Only One Element
if(Front == Rear)
{
    Front = Rear = -1;
}
else
{
    Front++;
}

cout << "\nData Deleted Successfully...\n\n";
}

void Peek()
{
    system("cls");

    cout << "Display Queue Module\n\n";

    if(Front == -1)
    {
        cout << "Queue is Empty...\n\n";
        return;
    }
}
```

```
}
```

```
cout << "Queue Elements are -- \n\n";
```

```
for(int i = Front; i <= Rear; i++)
```

```
{
```

```
    cout << QUEUE[i] << " -> ";
```

```
}
```

```
cout << "NULL\n\n";
```

```
}
```

```
void MainMenu()
```

```
{
```

```
    while(true)
```

```
    {
```

```
        system("cls");
```

```
        cout << "\tQUEUE OPERATIONS\n\n";
```

```
        cout << "1. Push\n";
```

```
        cout << "2. Pop\n";
```

```
        cout << "3. Display Queue\n";
```

```
        cout << "4. Exit\n\n";
```

```
        cout << "Choose Any Option -- ";
```



```
char c = getch();
```

```
switch(c)
```

```
{
```

```
    case '1':
```

```
        Push();
```

```
        break;
```

```
    case '2':
```

```
        Pop();
```

```
        break;
```

```
    case '3':
```

```
        Peek();
```

```
        break;
```

```
    case '4':
```

```
        exit(0);
```

```
    default:
```

```
        cout << "\nInvalid Choice...\n";
```

```
}
```

```
cout << "\nPress Any Key To Continue...";
```

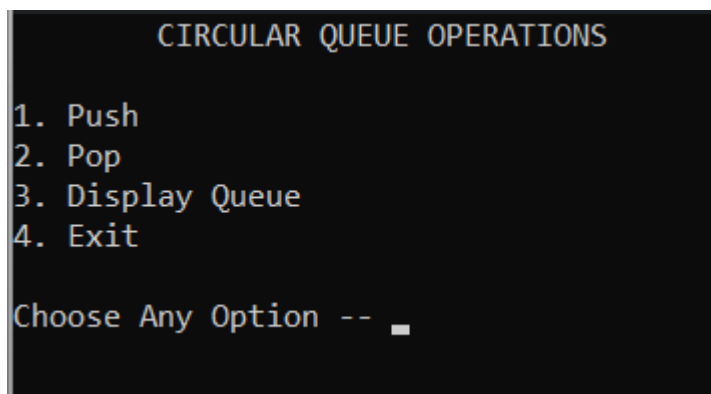
```
getch();
```

```
    }  
}  
};
```

```
int main()  
{  
    QueueOperation Q;  
    Q.MainMenu();  
  
    return 0;  
}
```

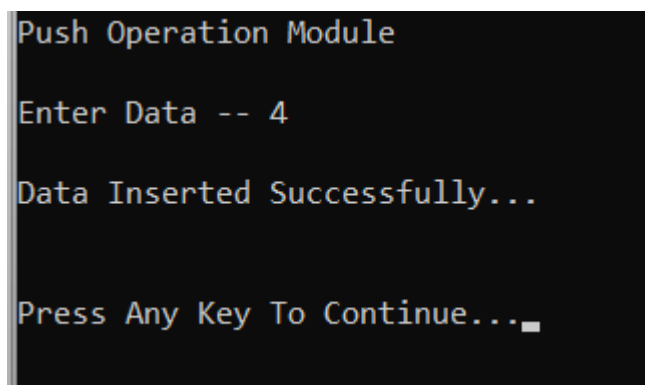
Output:

Menu



```
          CIRCULAR QUEUE OPERATIONS  
  
1. Push  
2. Pop  
3. Display Queue  
4. Exit  
  
Choose Any Option -- █
```

Push



```
Push Operation Module  
  
Enter Data -- 4  
  
Data Inserted Successfully...  
  
Press Any Key To Continue... █
```

Pop

```
Pop Operation Module

Deleted Element -- 4

Data Deleted Successfully...

Press Any Key To Continue...
```

Peek

```
Display Circular Queue Module

Queue Elements are --

4 -> 5 -> 8 -> NULL

Press Any Key To Continue..._
```

2.WAP to input size of Circular Queue from user and create a circular queue using array. And perform push, pop and traverse operations using menu driven programming and functions.

Code:

```
#include<iostream>

#include<conio.h>

using namespace std;

class CQueue
{
public:
    int CQUEUE[10];
};

class CQueueOperation : public CQueue
```

```

{
public:
    int Front,Rear,MaxSize;
    CQueueOperation()
    {
        Front = Rear = -1 ;
        cout<<"Enter Size of Queue -- ";
        cin>>MaxSize;
    }
    int Push()
    {
        system("cls");
        cout<<"Push Operation Module \n\n";
        int DATA , PushStatus=0;
        cout<<"Enter Data - ";
        cin>>DATA;
        if(Front == 0 && Rear == MaxSize-1){
            cout<<"Queue is Overflow...\n\n";
            return 0;
        }
        if(Front == -1 && Rear == -1){
            Front = Rear = 0;
            CQUEUE[Rear] = DATA;
            PushStatus = 1 ;
        }
        else if (Rear == MaxSize-1 && Front!= 0){

```

```

    Rear = 0;
    CQUEUE[Rear] = DATA;
    PushStatus = 1;
}
else {
    Rear = Rear + 1;
    CQUEUE[Rear] = DATA;
    PushStatus = 1;
}
if(PushStatus == 1)
    cout<<"\n\nData Successfully Inserted/Pushed..\n\n";
}

int Pop()
{
    system("cls");
    cout<<"Pop From Queue Module \n\n";
    if(Front == -1 && Rear == -1){
        cout<<"Queue is Underflow...\n\n";
        return 0;
    }
    if(Front == Rear){
        CQUEUE[Front] = 0;
        Front = Rear = -1;
    }else if( Front == MaxSize-1) {
        CQUEUE[Front] = 0;
        Front = 0;
    }
}

```

```

    }
else {
    CQUEUE[Front]= 0;
    Front = Front + 1;
}
cout<<"Data Deleted Successfully...\n\n";
}
int Peek()
{
    if(Front == -1){
        cout<<"Queue is Empty..\n\n";
        return 0;
    }
    cout<<"Queue Element are -- \n\n";
    for(int POS = Front ; 1 ;){
        cout<<POS<<" - "<<CQUEUE[POS]<<" -> ";
        if( POS == Rear){
            break;
        }
        POS = (POS = MaxSize-1 && POS != Rear)?0:++POS;
    }
    cout<<"\n\n";

}

void MainMenu()
{

```

```

system("cls");
cout<<"\tCircular Queue Operation -- \n\n";
cout<<"1.Push..\n";
cout<<"2.Pop..\n";
cout<<"3.Peek..\n";
cout<<"4.Exit..\n";
cout<<"Choose any one of these..\n\n";
char c = getch();
if(c == '1'){
    Push();
}else if (c == '2'){
    Pop();
}else if (c == '3'){
    Peek();
} else if (c == '4'){
    exit(0);
}
cout<<"Do you want to continue press 1 else any key...\n";
c = getch();
if(c == '1'){
    MainMenu();
}
}
};

int main()
{

```

```
CQueueOperation C;  
C.MainMenu();  
}
```

Output:

Menu

```
          QUEUE OPERATIONS  
  
1. Push  
2. Pop  
3. Display Queue  
4. Exit  
  
Choose Any Option --
```

Push

```
Push Operation Module  
  
Enter Data -- 5  
  
Data Inserted Successfully...  
  
Press Any Key To Continue...
```

Pop

```
Pop Operation Module  
  
Deleted Element -- 5  
  
Data Deleted Successfully...  
  
Press Any Key To Continue..._
```

Peek


```
Display Queue Module

Queue Elements are --

5 -> 8 -> 9 -> NULL

Press Any Key To Continue...
```

3.WAP to input size of Stack from user and create a stack using array. And perform push, pop and peek operations using menu driven programming and functions.

Code:

```
#include<iostream>
```

```
#include<conio.h>
```

```
using namespace std;
```

```
class Stack
```

```
{
```

```
public:
```

```
    int STACK[10];
```

```
};
```

```
class StackOperation : public Stack
```

```
{
```

```
public:
```

```
    int TOP, MaxSize;
```

```

StackOperation()
{
    TOP = -1;

    cout << "Enter Size of Stack (1-10) -- ";
    cin >> MaxSize;

    if(MaxSize > 10 || MaxSize <= 0)
    {
        cout << "Invalid Stack Size...";
        exit(0);
    }
}

void Push()
{
    system("cls");

    cout << "Push In Stack Module\n\n";

    // Overflow Condition
    if(TOP == MaxSize - 1)
    {
        cout << "Stack Overflow...\n\n";
        return;
    }
}

```

```
int DATA;
```

```
cout << "Enter Data -- ";
```

```
cin >> DATA;
```

```
TOP++;
```

```
STACK[TOP] = DATA;
```

```
cout << "\nData Pushed Successfully...\n\n";
```

```
}
```

```
void Pop()
```

```
{
```

```
system("cls");
```

```
cout << "Pop From Stack Module\n\n";
```

```
// Underflow Condition
```

```
if(TOP == -1)
```

```
{
```

```
    cout << "Stack Underflow...\n\n";
```

```
    return;
```

```
}
```

```
cout << "Deleted Element -- " << STACK[TOP] << endl;
```

```
STACK[TOP] = 0;
```

```
TOP--;
```

```
cout << "\nData Deleted Successfully...\n\n";
```

```
}
```

```
void Peek()
```

```
{
```

```
    system("cls");
```

```
    cout << "Display Stack Module\n\n";
```

```
    if(TOP == -1)
```

```
    {
```

```
        cout << "Stack is Empty...\n\n";
```

```
        return;
```

```
    }
```

```
    cout << "Stack Elements are -- \n\n";
```

```
    for(int i = TOP; i >= 0; i--)
```

```
    {
```

```
        cout << STACK[i] << " -> ";
```

```
}
```

```
cout << "NULL\n\n";
```

```
}
```

```
void MainMenu()
```

```
{
```

```
while(true)
```

```
{
```

```
system("cls");
```

```
cout << "\tSTACK OPERATIONS\n\n";
```

```
cout << "1. Push\n";
```

```
cout << "2. Pop\n";
```

```
cout << "3. Display Stack\n";
```

```
cout << "4. Exit\n\n";
```

```
cout << "Choose Any Option -- ";
```

```
char c = getch();
```

```
switch(c)
```

```
{
```

```
case '1':
```

```
    Push();
```

```
        break;

    case '2':
        Pop();
        break;

    case '3':
        Peek();
        break;

    case '4':
        exit(0);

    default:
        cout << "\nInvalid Choice...\n";
    }

    cout << "\nPress Any Key To Continue...";
    getch();
}

};

int main()
{
    StackOperation S;
```

```
S.MainMenu();

return 0;
}
```

Output:

Menu

```
STACK OPERATIONS
1. Push
2. Pop
3. Display Stack
4. Exit
Choose Any Option -- █
```

Push

```
Push In Stack Module
Enter Data -- 4
Data Pushed Successfully...
Press Any Key To Continue...
```

Pop

```
Pop From Stack Module
Deleted Element -- 8
Data Deleted Successfully...
Press Any Key To Continue... █
```

Peek

```
Display Stack Module
```

```
Stack Elements are --
```

```
8 -> 6 -> 4 -> NULL
```

```
Press Any Key To Continue...■
```




Thank You

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